

```
Select count(id) as Total_Loan_Applications from financial_loan;
```

	Total_Loan_Applications
1	38576

```
Select count(id) as MTD_Total_Loan_Applications from financial_loan
where MONTH(issue_date)=12;
--as this data is from 2021 and curdate will have 2026 so we assume the latest date and month for calculating month to date that is december--
```

	MTD_Total_Loan_Applications
1	4314

```
SELECT
    current_month - previous_month AS MoM_Change,
    ROUND(
        (current_month - previous_month) * 100.0 / previous_month,
        2
    ) AS MoM_Percentage
FROM (
    SELECT
        COUNT(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 12
            THEN id
        END) AS current_month,
        COUNT(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 11
            THEN id
        END) AS previous_month
    FROM financial_loan
) x;
```

	MoM_Change	MoM_Percentage
1	279	6.910000000000

```
Select SUM(loan_amount) as total_funded_Amount from financial_loan;
```

	total_funded_Amount
1	435757075

```
Select SUM(loan_amount) as total_month_to_date_funded_Amount from financial_loan where month(issue_date)=12 and year(issue_date)=2021;
```

	total_month_to_date_funded_Amount
1	53981425

```

SELECT
    current_month - previous_month AS MoM_Change,
    ROUND(
        (current_month - previous_month) * 100.0 / previous_month,
        2
    ) AS MoM_Percentage
FROM (
    SELECT
        SUM(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 12
            THEN loan_Amount
        END) AS current_month,

        SUM(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 11
            THEN loan_amount
        END) AS previous_month
    FROM financial_loan
) x;

```

	MoM_Change	MoM_Percentage
1	6226600	13.0400000000000

```

Select SUM(total_payment) as total_received_Amount from financial_loan;

```

	total_received_Amount
1	473070933

```

Select SUM(total_payment) as total_month_to_date_received_Amount from financial_loan where month(issue_date)=12 and year(issue_date)=2021;

```

	total_month_to_date_received_Amount
1	58074380

```

SELECT
    current_month - previous_month AS MoM_Change,
    ROUND(
        (current_month - previous_month) * 100.0 / previous_month,
        2
    ) AS MoM_Percentage
FROM (
    SELECT
        SUM(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 12
            THEN total_payment
        END) AS current_month,

        SUM(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 11
            THEN total_payment
        END) AS previous_month
    FROM financial_loan
) x;

```

	MoM_Change	MoM_Percentage
1	7942350	15.8400000000000

```

Select ROUND(AVG(int_rate),4) * 100 as Avg_Interest_Rate from financial_loan ;

```

	Avg_Interest_Rate
1	12.05

```

Select ROUND(AVG(int_rate),4) * 100 as total_month_to_date_funded_Amount from financial_loan where month(issue_date)=12 and year(issue_date)=2021;

```

	total_month_to_date_funded_Amount
1	12.36

```

SELECT
    current_month - previous_month AS MoM_Change,
    ROUND(
        (current_month - previous_month) * 100.0 / previous_month,
        2
    ) AS MoM_Percentage
FROM (
    SELECT
        ROUND(AVG(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 12
            THEN int_rate
        END), 4) AS current_month,

        ROUND(AVG(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 11
            THEN int_rate
        END), 4) AS previous_month

    FROM financial_loan
) x;

```

	MoM_Change	MoM_Percentage
1	0.0042	3.52

```

Select ROUND(AVG(dti),4) * 100 as Avg_DTI_Rate from financial_loan ;

```

	Avg_DTI_Rate
1	13.33

```

Select ROUND(AVG(dti),4) * 100 as Avg_month_to_date_DTI_Rate from financial_loan where month(issue_date)=12 and year(issue_date)=2021;

```

	Avg_month_to_date_DTI_Rate
1	13.67

```

SELECT
    current_month - previous_month AS MoM_Change,
    ROUND(
        (current_month - previous_month) * 100.0 / previous_month,
        2
    ) AS MoM_Percentage
FROM (
    SELECT
        ROUND(AVG(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 12
            THEN dti
        END), 4) AS current_month,

        ROUND(AVG(CASE
            WHEN YEAR(issue_date) = 2021
            AND MONTH(issue_date) = 11
            THEN dti
        END), 4) AS previous_month

    FROM financial_loan
) x;

```

	MoM_Change	MoM_Percentage
1	0.003699999999999998	2.78

```

--Good loan application percentage--
select
    count(case when loan_status = 'Fully Paid' or loan_status = 'Current' then id end)*100
    /
    count(id)
    as good_loan_percentage
from financial_loan;

```

	good_loan_percentage
1	86

```

select count(id) as good_loan_applications from financial_loan where loan_status = 'Fully Paid' or loan_status = 'Current' ;

```

	good_loan_applications
1	33243

```

select sum(loan_amount) as good_loan_funded_amount from financial_loan
where loan_status = 'Fully paid' or loan_status = 'Current';

```

	good_loan_funded_amount
1	370224850

```
select sum(total_payment) as good_loan_received_amount from financial_loan
where loan_status = 'Fully paid' or loan_status = 'Current';
```

	good_loan_received_amount
1	435786170

--bad loan application percentage--

```
select
    count(case when loan_status = 'Charged off' then id end)*100 /
    count(id)
    as bad_loan_percentage
from financial_loan;
```

	bad_loan_percentage
1	13

```
select count(id) as bad_loan_applications from financial_loan where loan_status = 'charged off';
```

	bad_loan_applications
1	5333

```
select sum(loan_amount) as bad_loan_funded_amount from financial_loan
where loan_status = 'Charged off';
```

	bad_loan_funded_amount
1	65532225

```
select sum(total_payment) as bad_loan_received_amount from financial_loan
where loan_status = 'Charged off';
```

	bad_loan_received_amount
1	37284763

```
select loan_status,
    count(id) as Total_loan_applications,
    sum(total_payment) as Total_amount_received,
    sum(loan_amount) as Total_Funded_Amount,
    AVG(int_rate*100) as Interest_Rate,
    AVG(dti*100) as DTI
from financial_loan
group by loan_status;
```

	loan_status	Total_loan_applications	Total_amount_received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

```

select loan_status,
       count(id) as MTD_Total_loan_applications,
       sum(total_payment) as MTD_Total_amount_received,
       sum(loan_amount) as MTD_Total_Funded_Amount,
       AVG(int_rate*100) as MTD_Interest_Rate,
       AVG(dti*100) as MTD_DTI
from financial_loan
where month(issue_date)=12
group by loan_status;

```

	loan_status	MTD_Total_loan_applications	MTD_Total_amount_received	MTD_Total_Funded_Amount	MTD_Interest_Rate	MTD_DTI
1	Fully Paid	3452	47815851	41302025	11.7703332126503	13.3745451938834
2	Charged Off	649	5324211	8732775	14.2535593842826	14.7357627137411
3	Current	213	4934318	3946625	16.0667136731282	15.1206103620281

--Monthly Trends by issue date--

```

select
       month(issue_date) as month_number,
       datename(month,issue_date) as month_name,
       count(id) as total_loan_applications,
       sum(loan_amount) as total_funded_amount,
       sum(total_payment) as total_received_amount
from financial_loan
group by month(issue_date),datename(month,issue_date)
order by month(issue_date);

```

	month_number	month_name	total_loan_applications	total_funded_amount	total_received_amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

```
--regional analysis by state--
select
    address_state,
    count(id) as total_loan_applications,
    sum(loan_amount) as total_funded_amount,
    sum(total_payment) as total_received_amount
from financial_loan
group by address_state
order by address_state;
```

	address_state	total_loan_applications	total_funded_amount	total_received_amount
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1210	15051000	16676270

```
--loan term analysis--
select
    term,
    count(id) as total_loan_applications,
    sum(loan_amount) as total_funded_amount,
    sum(total_payment) as total_received_amount
from financial_loan
group by term
order by term;
```


	term	total_loan_applications	total_funded_amount	total_received_amount
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

--employee length analysis--

```
select
    emp_length,
    count(id) as total_loan_applications,
    sum(loan_amount) as total_funded_amount,
    sum(total_payment) as total_received_amount
from financial_loan
group by emp_length
order by emp_length;
```

	emp_length	total_loan_applications	total_funded_amount	total_received_amount
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

--loan purpose breakdown--

```
select
    purpose,
    count(id) as total_loan_applications,
    sum(loan_amount) as total_funded_amount,
    sum(total_payment) as total_received_amount
from financial_loan
group by purpose
order by count(id);
```

	purpose	total_loan_applications	total_funded_amount	total_received_amount
1	renewable_energy	94	845750	898931
2	educational	315	2161650	2248380
3	vacation	352	1967950	2116738
4	house	366	4824925	5185538
5	moving	559	3748125	3999899
6	medical	667	5533225	5851372
7	wedding	928	9225800	10266856
8	car	1497	10223575	11324914
9	small business	1776	24123100	23814817
10	major purchase	2110	17251600	18676927
11	home improvement	2876	33350775	36380930
12	other	3824	31155750	33289676
13	credit card	4998	58885175	65214084
14	Debt consolidation	18214	232459675	253801871

--home ownership analysis--

```
select
    home_ownership,
    count(id) as total_loan_applications,
    sum(loan_amount) as total_funded_amount,
    sum(total_payment) as total_received_amount
from financial_loan
group by home_ownership
order by count(id);
```

	home_ownership	total_loan_applications	total_funded_amount	total_received_amount
1	NONE	3	16800	19053
2	OTHER	98	1044975	1025257
3	OWN	2838	29597675	31729129
4	MORTGAGE	17198	219329150	238474438
5	RENT	18439	185768475	201823056

Untitled - Power BI Desktop

File Home Help Table tools

Name Date Table

Structure

1 Date Table - CALENDAR(MIN(financial_loan[issue_date]),MAX(financial_loan[issue_date]))

Date

01-01-2021 00:00:00
02-01-2021 00:00:00
03-01-2021 00:00:00
04-01-2021 00:00:00
05-01-2021 00:00:00
06-01-2021 00:00:00
07-01-2021 00:00:00
08-01-2021 00:00:00
09-01-2021 00:00:00
10-01-2021 00:00:00
11-01-2021 00:00:00
12-01-2021 00:00:00
13-01-2021 00:00:00
14-01-2021 00:00:00
15-01-2021 00:00:00
16-01-2021 00:00:00
17-01-2021 00:00:00
18-01-2021 00:00:00
19-01-2021 00:00:00
20-01-2021 00:00:00
21-01-2021 00:00:00
22-01-2021 00:00:00
23-01-2021 00:00:00
24-01-2021 00:00:00
25-01-2021 00:00:00
26-01-2021 00:00:00
27-01-2021 00:00:00
28-01-2021 00:00:00
29-01-2021 00:00:00
30-01-2021 00:00:00
31-01-2021 00:00:00
01-02-2021 00:00:00

Table: Date Table (346 rows)

Update available (click to download)

Table tools

Manage relationships Relationships

New measure measure column Calculations

Quick New table

New table

Mark as date table Calendars

Data

Search

Date Table

financial_loan

address_state

annual_income

application_type

dti

emp_length

emp_title

grade

home_ownership

id

installment

int_rate

issue_date

last_credit_pull_date

last_payment_date

loan_amount

loan_status

member_id

next_payment_date

purpose

sub_grade

term

total_acc

total_payment

verification_status

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Month

Data type Text

Format Text

Summarization Don't summarize

Data category Uncategorized

Sort by column

Data groups

Manage relationships Relationships

New column Calculations

Structure

1 Month = FORMAT('Date Table'[Date], "mmmm")

Date

Month

01-01-2021 00:00:00 January
02-01-2021 00:00:00 January
03-01-2021 00:00:00 January
04-01-2021 00:00:00 January
05-01-2021 00:00:00 January
06-01-2021 00:00:00 January
07-01-2021 00:00:00 January
08-01-2021 00:00:00 January
09-01-2021 00:00:00 January
10-01-2021 00:00:00 January
11-01-2021 00:00:00 January
12-01-2021 00:00:00 January
13-01-2021 00:00:00 January
14-01-2021 00:00:00 January
15-01-2021 00:00:00 January
16-01-2021 00:00:00 January
17-01-2021 00:00:00 January
18-01-2021 00:00:00 January
19-01-2021 00:00:00 January
20-01-2021 00:00:00 January
21-01-2021 00:00:00 January
22-01-2021 00:00:00 January
23-01-2021 00:00:00 January
24-01-2021 00:00:00 January
25-01-2021 00:00:00 January
26-01-2021 00:00:00 January
27-01-2021 00:00:00 January
28-01-2021 00:00:00 January
29-01-2021 00:00:00 January
30-01-2021 00:00:00 January
31-01-2021 00:00:00 January
01-02-2021 00:00:00 February

Table: Date Table (346 rows) Column: Month (12 distinct values)

Update available (click to download)

Column tools

Format Text

Summarization Don't summarize

Data category Uncategorized

Sort by column

Data groups

Manage relationships Relationships

New column Calculations

Data

Search

Date Table

Date

Month

financial_loan

address_state

annual_income

application_type

dti

emp_length

emp_title

grade

home_ownership

id

installment

int_rate

issue_date

last_credit_pull_date

last_payment_date

loan_amount

loan_status

member_id

next_payment_date

purpose

sub_grade

term

total_acc

total payment